PURNELL ENERGY SERVICES

QF-9

Pressure Test Report

Customer: Arc Resources

Job Number: 2024-062A

Project: 10-16 Sales Gas Rerouting (10-16)

Site/Location: 10-16-062-04 W6M

Piping or Line Tested: See Page 2

Test Package Number: 1

Construction Code:	ASME B31.3	**Location of Test:**	Purnell Shop
Flange Rating:	600 #	**Test Duration:**	15 Mins
Design Pressure:	10,200 kPa (1,479 psi)	**Test Medium:**	Water
Test Pressure Min:	15,315 kPa (2,221 psi)	**Test Pressure Max:**	15,596 kPa (2,262 psi)

Ambient Temperature:

Test Temperature:

Dead Weight Pressure Range:	0 to 68,948 kPa (0 to 10,000 psi)	**Dead Weight Serial Number:**	PES: DDW-7
Test Recorder Pressure Range:	0 to 68,948 kPa (0 to 10,000 psi)	**Test Recorder Serial Number:**	PES: PT-1
Test Recorder Pressure Range:	N/A	**Test Recorder Serial Number:**	N/A
Gauge Pressure Range:	N/A	**Gauge Serial Number:**	N/A

Purnell Representative:		
Cameron Janhsen		April 29, 2024
Name (print or type)	**Signature**	**Date (m/d/y)**
Customer Representative:		
Robyn Maloney		April 29, 2024
Name (print or type)	**Signature**	**Date (m/d/y)**

Line Number:	Revision Number:	ISO Number:	Control Number:
273.0-NG-CLS-720522-38H	0	273.0-NG-CLS-720522-38H-A	1
273.0-NG-CLS-720522-38H	0	273.0-NG-CLS-720522-38H-B	2
60.3-NG-CLS-720540-25H	0	60.3-NG-CLS-720540-25H-A	3
60.3-NG-CLS-720540-25H	0	60.3-NG-CLS-720540-25H-B	4
60.3-NG-CLS-720540-25H	0	60.3-NG-CLS-720540-25H-C	5
168.3-SP-CLS-725310-38H	0	168.3-SP-CLS-725310-38H-A	6
168.3-SP-CLS-725310-38H	0	168.3-SP-CLS-725310-38H-B	7
168.3-SP-CLS-725310-38H	0	168.3-SP-CLS-725310-38H-C	8
168.3-NG-CLS-725311-38H	0	168.3-NG-CLS-725311-38H-A	9
168.3-NG-CLS-725311-38H	1	168.3-NG-CLS-725311-38H-B	10
60.3-NG-CLS-730218-25H	0	60.3-NG-CLS-730218-25H-A	11
60.3-NG-CLS-730218-25H	0	60.3-NG-CLS-730218-25H-B	12
60.3-NG-CLS-730218-25H	0	60.3-NG-CLS-730218-25H-C	13
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-A	14
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-B	15
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-C	16
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-D	17
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-E	18
323.8-NG-CLS-730512-51H	0	323.8-NG-CLS-730512-51H-F	19
60.3-NG-CLS-730513-25H	0	60.3-NG-CLS-730513-25H	20
33.4-NG-CLS-730219-25H	0	33.4-NG-CLS-730219-25H	23
ASSORTED THREADED NIPPLES, PIPE, VALVES, AND FITTINGS			N/A

BURKE CALIBRATION
11455 98th Ave
Grande Prairie, Alberta T8V 5S5
(780) 897-5869

CERTIFICATE OF CALIBRATION

35706 DDW-007

Report No.: BCL2022-C13455
Model/Gauge Uncertainty: +/- ASME 3A of span (0.25%)

Manufacturer	Model	Serial Number	Calibration Date	Expiration Date
CALSCAN	FALCON-10K	35706	2023-11-28	2024-05-27

As Received Condition: In tolerance As Left Condition: In tolerance

AS RECEIVED TEST RESULTS 10000 PSI

Reference Reading	Gauge Reading	Allowable Tolerance	Difference	Diff. % of Reading		Condition
0	0	25	0			Pass
1999	2000	25	1	0.05%		Pass
3999	4000	25	1	0.03%		Pass
5997	6000	25	3	0.05%		Pass
7998	8000	25	2	0.03%		Pass
9996	10000	25	4	0.04%		Pass

AS LEFT TEST RESULTS 10000 PSI

Reference Reading	Gauge Reading	Allowable Tolerance	Difference	Diff. % of Reading		Condition
0	0	25	0			Pass
1999	2000	25	1	0.05%		Pass
3999	4000	25	1	0.03%		Pass
5997	6000	25	3	0.05%		Pass
7998	8000	25	2	0.03%		Pass
9996	10000	25	4	0.04%		Pass

Laboratory ambient conditions throughout this calibration were:

Tempurature: 20 to 24° C
Humidity: 30 to 35% RH
Pressure: 100 to 103 kPa

Calibration Performed by:

Reference Standards used in this calibration are traceable to the National Institute of Standards and Technology of the United States, through the following report numbers:

Manufacturer	Model	Serial Number	Report Number	Due Date	Reference Uncertainty
Crystal Engineering	10KPSIXP2I	545487	69253	2024-01-11	0-20% of FS: ±(0.02% of FS); 20%-100% of FS: ±(0.1% of Rdg)

All calibrations are performed in a controlled environment by qualified personnel using instrumentation and methods which guarantee that specifications claimed are reliable. Calibrations conform to ANSI/NCSL Z540-1-1994, MIL-STD 45662A, 10CFR21 and 10CFR50

20170615

CONTROLTECH

Transmitter Calibration

COMPANY (REQUIRED)	Purnell	DATE (REQUIRED)	April 16 2024 next cal April 16 2025	
LSD	Test Station 1	TECHNICIAN(S)	Kelsey Fraser	

CALIBRATION EQUIPMENT

Manufacturer	Model	Serial (Gauge) Number	Model/Guage Uncertainty	Calibration Data	Expiration Date
Fluke	7003G1	2377022'	+/- ASME 4A of span (0.1%)	2024-03-28	2025-03-28
Fluke	7003G1	4811301'	+/- ASME 4A of span (0.1%)	2024-03-28	2025-03-28
Fluke	1551A EX	4199009'	+/- ASME 4A of span (0.05%)	2023-10-27	2024-10-27

Equipment Type (REQUIRED)	TRANSMITTER - STATIC PRESS		
TAG (REQUIRED)	PT-1	MODEL NUMBER	2051
MANUFACTURER NAME	Rosemount	SERIAL NO.	245897
%	Applied P.V.	As Found	As Left
0	0	3.12	0
50	5000	5003.71	4999.94
100	10000	10003	9999.92
Bypass	Yes	Bypass Removed?	Yes
COMMENTS:			

Equipment Type (REQUIRED)	TRANSMITTER - TEMP		
TAG (REQUIRED)	TT-1	MODEL NUMBER	344
MANUFACTURER NAME	Rosemount	SERIAL NO.	118261
%	Applied P.V.	As Found	As Left
0	0	0.31	0.31
50	20	20.3	20.4
100	50	50.2	50.2
Bypass	Yes	Bypass Removed?	Yes
COMMENTS:			

Equipment Type (REQUIRED)			
TAG (REQUIRED)		MODEL NUMBER	
MANUFACTURER NAME		SERIAL NO.	
%	Applied P.V.	As Found	As Left
0			
50			
100			
Bypass		Bypass Removed?	
COMMENTS:			

Equipment Type (REQUIRED)			
TAG (REQUIRED)		MODEL NUMBER	
MANUFACTURER NAME		SERIAL NO.	
%	Applied P.V.	As Found	As Left
0			
50			
100			
Bypass		Bypass Removed?	
COMMENTS:			

Equipment Type (REQUIRED)			
TAG (REQUIRED)		MODEL NUMBER	
MANUFACTURER NAME		SERIAL NO.	
%	Applied P.V.	As Found	As Left
0			
50			
100			
Bypass		Bypass Removed?	
COMMENTS:			

Equipment Type (REQUIRED)			
TAG (REQUIRED)		MODEL NUMBER	
MANUFACTURER NAME		SERIAL NO.	
%	Applied P.V.	As Found	As Left
0			
50			
100			
Bypass		Bypass Removed?	
COMMENTS:			

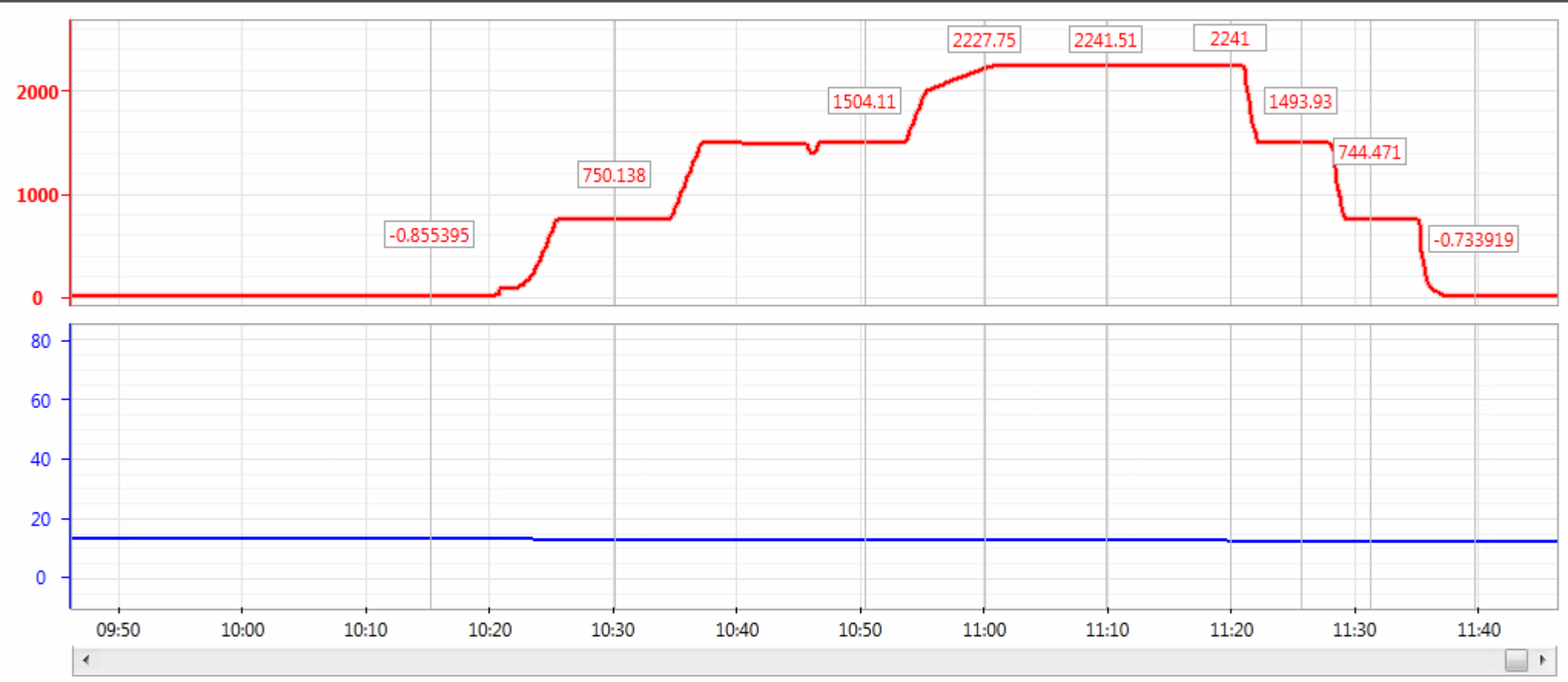
VTScadaLIGHT

Historical Data Viewer

11:46 AM Apr 29

Logon

Test Station 1 | Tag Selection | Export | 2 Hours | Time Span | Pan & Zoom | View | Note



Pen	Name	Description	Value	Minimum	Maximum	Average	Starts	On Time
	....\PT-1	Test Station 1 Pressur		-1.91 PSI	2,243 PSI	934 PSI		
	....\TT-1	Test Station 1 Tempe		11.7 degC	12.9 degC	12.3 degC		

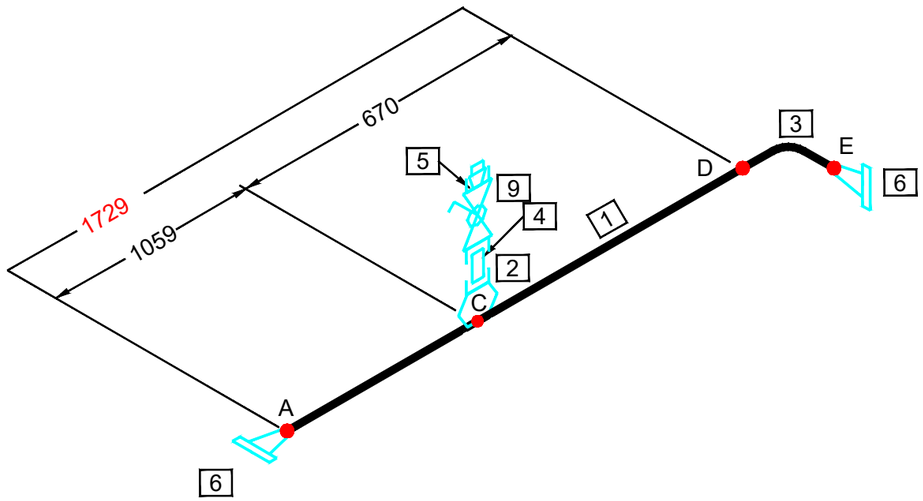
Plot | Grid | Notes

Historical Data Viewer

Test Station 1 | Tag Selection | Export | 2 Hours | Time Span | Averages | 5 Minutes | Data Mode & Time Per Record | Note

Time	Hydro Testing\PLC\TS1\TS1 MDBUS\PT-1 Test Station 1 Pressure	Hydro Testing\PLC\TS1\TS1 MDBUS\TT-1 Test Station 1 Temperature
09:45	10	12.67
09:50	-1	12.71
09:55	-1	12.76
10:00	-1	12.79
10:05	-1	12.83
10:10	-1	12.85
10:15	-1	12.88
10:20	173	12.74
10:25	745	12.23
10:30	752	12.34
10:35	1355	12.18
10:40	1483	12.23
10:45	1485	12.28
10:50	1556	12.33
10:55	2098	12.20
11:00	2240	12.28
11:05	2241	12.28
11:10	2241	12.20
11:15	2241	12.11
11:20	1720	11.94
11:25	1284	11.82
11:30	745	11.76
11:35	95	11.77
11:40	-1	11.76
11:45	-1	11.66

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2198	1	10	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	1729 mm
			FITTINGS	
51597	2	10X1	TOL, CL6000, A350-LF2 CL1, NACE	2
533B	3	10	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
I66	4	1	NIPPLE, SMLS, A333-6, NACE, 100mm LG, TBE	2
			(100.000 MM LG)	
YB0206	5	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	2
			FLANGES	
134005	6	10	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	2
			GASKETS	
	7	10	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB"	2
			CGI, 304L OR 316L SS WINDINGS AND INNER RING,	
			FLEXIBLE GRAPHITE FILLER, ASME B16.20	
			BOLTS	
	8	1.1/4	(16) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD)	2
			ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (220.000	
			MM LG)	
			VALVES / IN-LINE ITEMS	
231008	9	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316	2
			SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN,	
			FLOATING, LEAVER OP., NACE, 30BVTT4H	

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	10"	BW	JI	XF-53			DM
C	1"	LET	JI		MT-9		DM
D	10"	BW	JI	XF-52			DM
E	10"	BW	JI	XF-51		XF-51	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

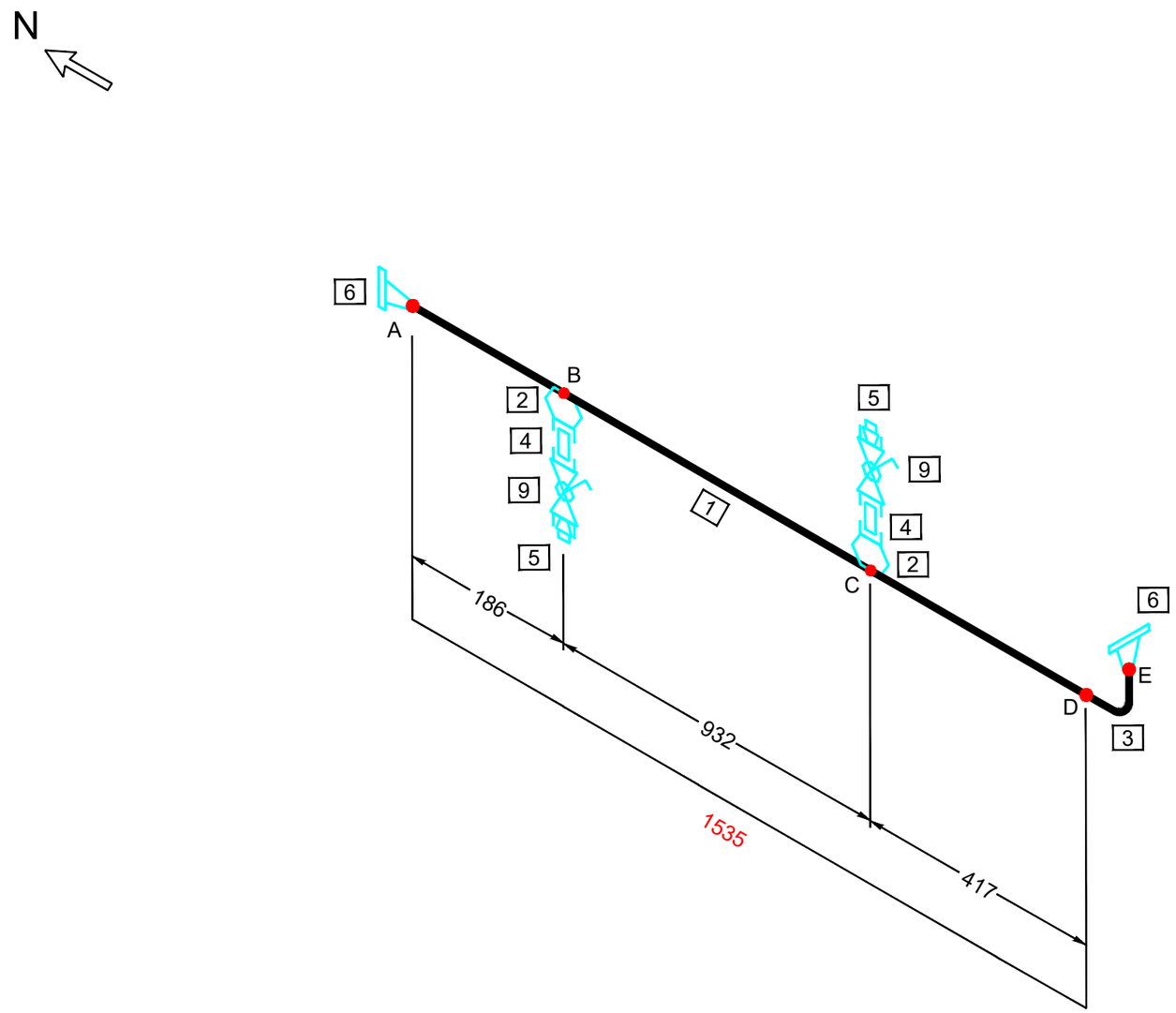
SPOOL NUMBER: 273.0-NG-CLS-720522-38H-A

REV: 0

CONTROL NUMBER: 1

ARC RESOURCES LTD.

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BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2198	1	10	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	1535 mm
			FITTINGS	
51597	2	10X1	TOL, CL6000, A350-LF2 CL1, NACE	2
533B	3	10	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
I66	4	1	NIPPLE, SMLS, A333-6, NACE, 100mm LG, TBE	2
YB0206	5	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	2
			FLANGES	
22/36388	6	10	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	2
			GASKETS	
	7	10	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB"	2
			CGI, 304L OR 316L SS WINDINGS AND INNER RING,	
			FLEXIBLE GRAPHITE FILLER, ASME B16.20	
			BOLTS	
	8	1.1/4	(16) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD)	2
			ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (220.000	
			MM LG)	
			VALVES / IN-LINE ITEMS	
231008	9	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316	2
			SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN,	
			FLOATING, LEAVER OP., NACE, 30BVTT4H	

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	10"	BW	JI	XF-3		XF-3	DM
B	1"	LET	JI		MT-1		DM
C	1"	LET	JI		MT-2		DM
D	10"	BW	JI	XF-4			DM
E	10"	BW	JI	XF-5			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.					
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

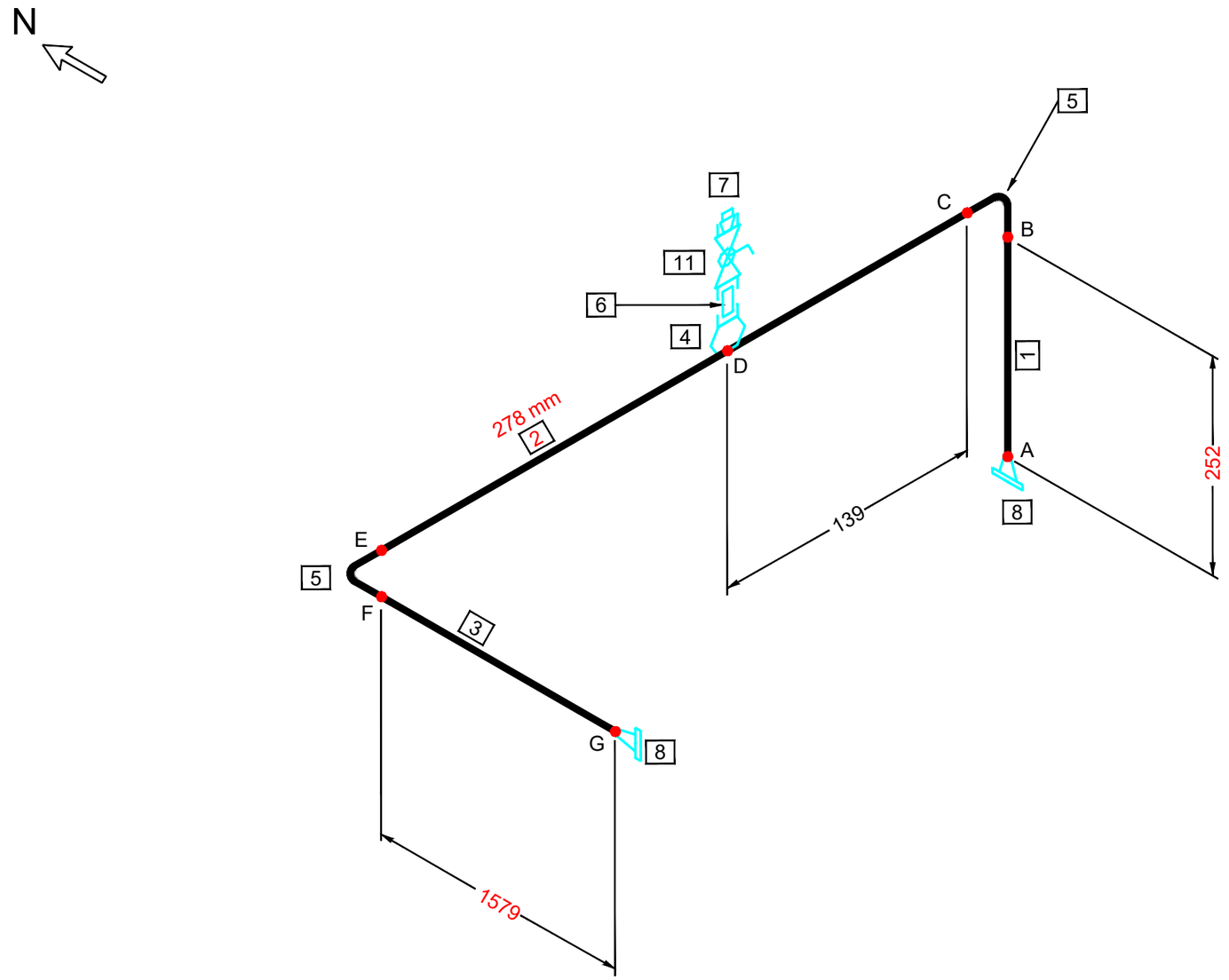
SPOOL NUMBER: 273.0-NG-CLS-720522-38H-B

ARC RESOURCES LTD.

PURNELL

REV: 0

CONTROL NUMBER 2



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	252 mm
AG2073	2	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	278 mm
AG2073	3	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	1579 mm
			FITTINGS	
51597	4	2X1	TOL, CL6000, A350-LF2 CL1, NACE	1
07VPY, 981B	5	2	ELBOW, 90 LR, SCH. 160, BW, A420-WPL6, NACE	2
I66	6	1	NIPPLE, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	1
YB0206	7	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
			FLANGES	
0618	8	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			GASKETS	
	9	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	10	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1
			VALVES / IN-LINE ITEMS	
231008	11	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-18		XF-18	DM
B	2"	BW	GC	XF-19			DM
C	2"	BW	GC	XF-20			DM
D	1"	LET	GC		MT-4		DM
E	2"	BW	GC	XF-21			DM
F	2"	BW	GC	XF-22			DM
G	2"	BW	GC	XF-23			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	25H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

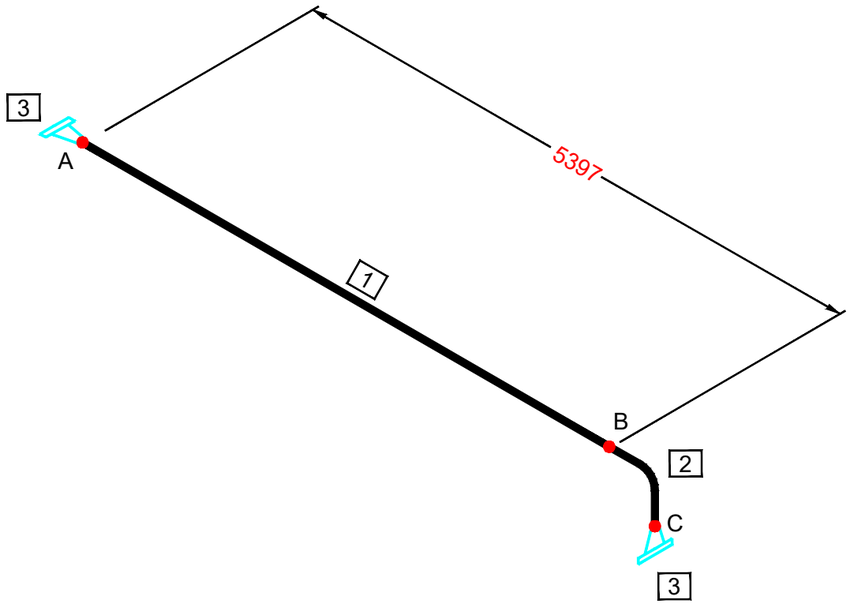
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-720540-25H-A

REV: 0

CONTROL NUMBER: 3

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BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	5397 mm
			FITTINGS	
07VPY	2	2	ELBOW, 90 LR, SCH. 160, BW, A420-WPL6, NACE	1
			FLANGES	
0618	3	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			GASKETS	
	4	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	5	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-45		XF-45	DM
B	2"	BW	GC	XF-46			DM
C	2"	BW	GC	XF-47			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	25H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-720540-25H-B

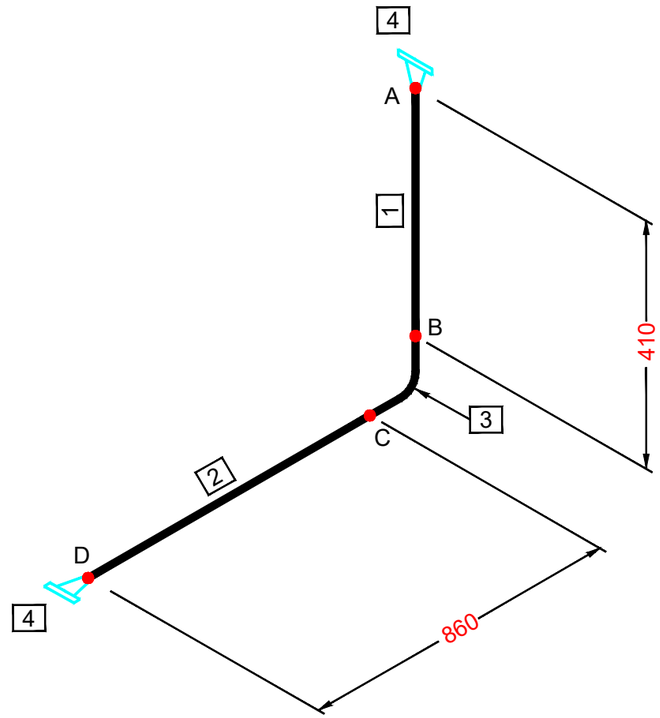
REV: 0

CONTROL NUMBER: 4

ARC RESOURCES LTD.

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BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2037	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	410 mm
AG2037	2	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	860 mm
			FITTINGS	
981B	3	2	ELBOW, 90 LR, SCH. 160, BW, A420-WPL6, NACE	1
			FLANGES	
0618	4	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			GASKETS	
	5	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	6	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-39		XF-39	DM
B	2"	BW	GC	XF-40			DM
C	2"	BW	GC	XF-41			DM
D	2"	BW	GC	XF-42			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	4	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	8.0	X-RAY	100%	INSULATION SPEC	25H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-720540-25H-C

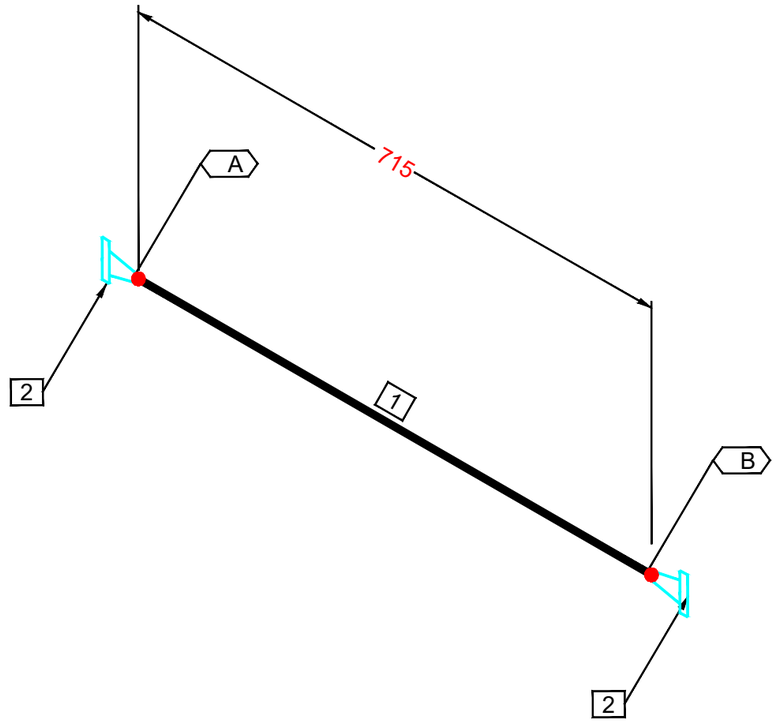
REV: 0

CONTROL NUMBER: 5

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
	1	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	715 mm
			FLANGES	
	2	8	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	2

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	8"	BW	GC	XF-55			DM
B	8"	BW	GC	XF-54		XF-54	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	2	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	16.0	X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

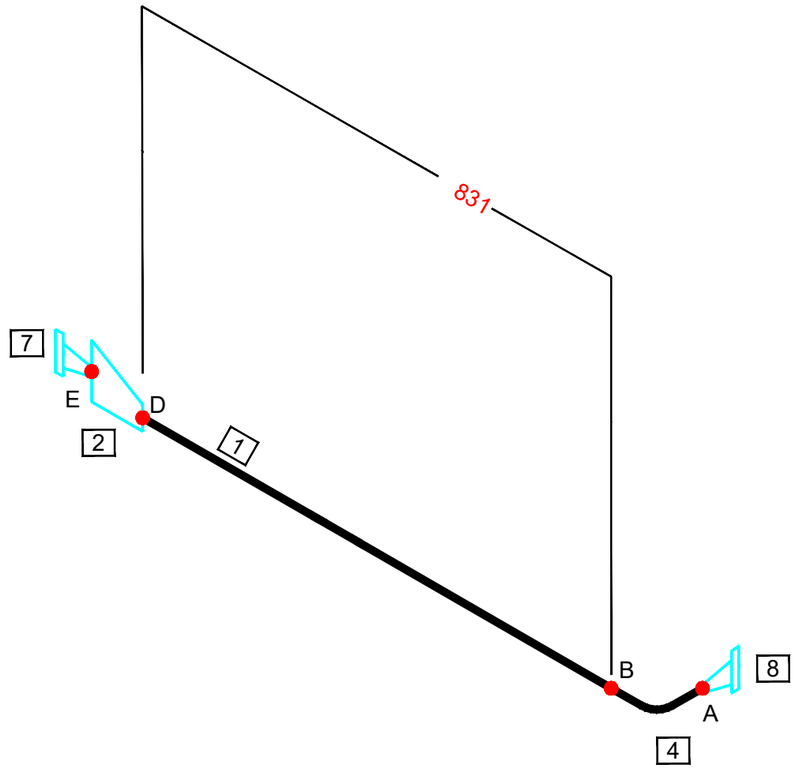
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 168.3-SP-CLS-725310-38H-A

REV: 0

CONTROL NUMBER: 6

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2266	1	6	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	831 mm
			FITTINGS	
9XX	2	8X6	REDUCER, ECC., SCH. 80 x SCH. 80, BW, A420-WPL6, NACE	1
NEED HT#	3	6X1	TOL, CL6000, A350-LF2 CL1, NACE	1
6XX	4	6	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
NEED HT#	5	1	NIPPLE, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	1
NEED HT#	6	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
			FLANGES	
2304	7	8	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
B2A08425	8	6	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
			GASKETS	
	9	6	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	10	1	(12) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (175.000 MM LG)	1
			VALVES / IN-LINE ITEMS	
	11	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	6"	BW	DJM	XF-9			DM
B	6"	BW	DJM	XF-8			DM
							DM
D	6"	BW	DJM	XF-7			DM
E	8"	BW	DJM	XF-6		XF-6	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.					
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 168.3-SP-CLS-725310-38H-B

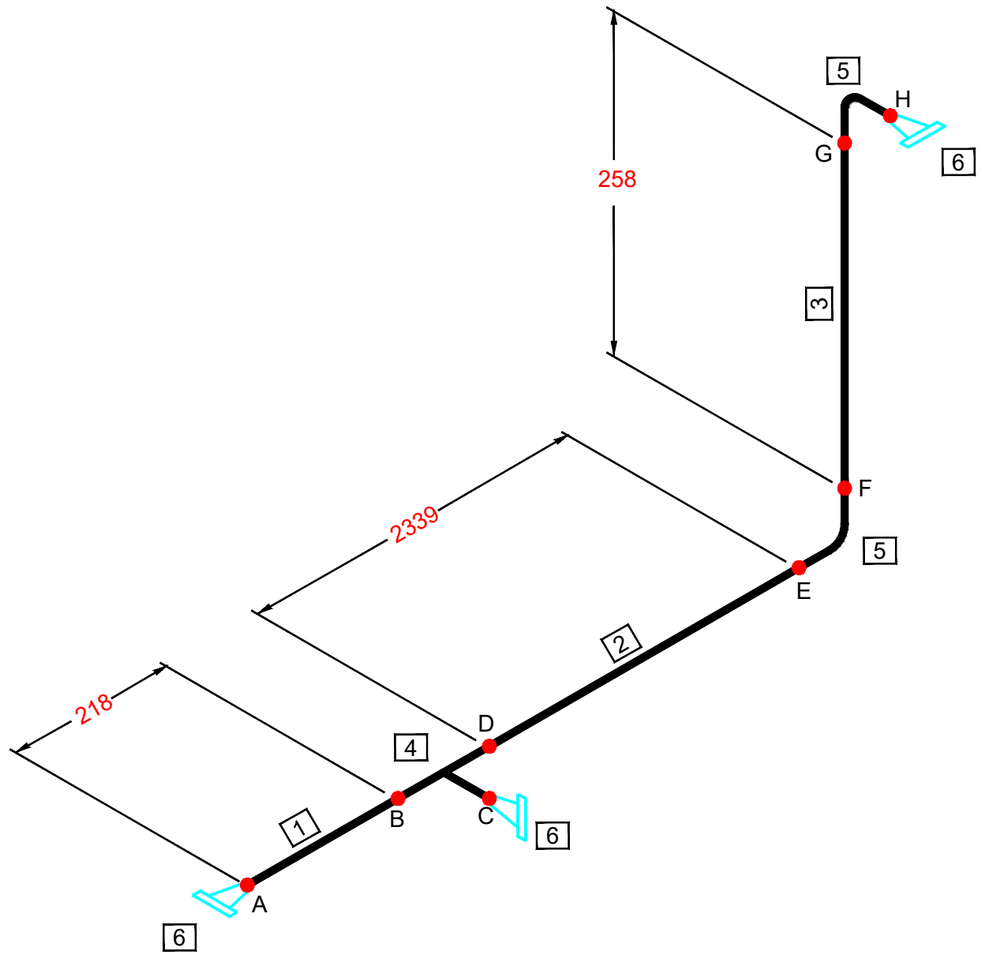
REV: 0

CONTROL NUMBER: 7

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2266	1	6	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	218 mm
AG2266	2	6	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	2339 mm
AG2266	3	6	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	258 mm
			FITTINGS	
866	4	6X6	TEE, STR., SCH. 80, BW, A420-WPL6, NACE	1
6XX	5	6	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	2
			FLANGES	
B2A08425	6	6	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	3

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	6"	BW	DJM	XF-10		XF-10	DM
B	6"	BW	DJM	XF-11			DM
C	6"	BW	DJM	XF-12			DM
D	6"	BW	DJM	XF-13			DM
E	6"	BW	DJM	XF-14			DM
F	6"	BW	DJM	XF-15			DM
G	6"	BW	DJM	XF-16			DM
H	6"	BW	DJM	XF-17			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

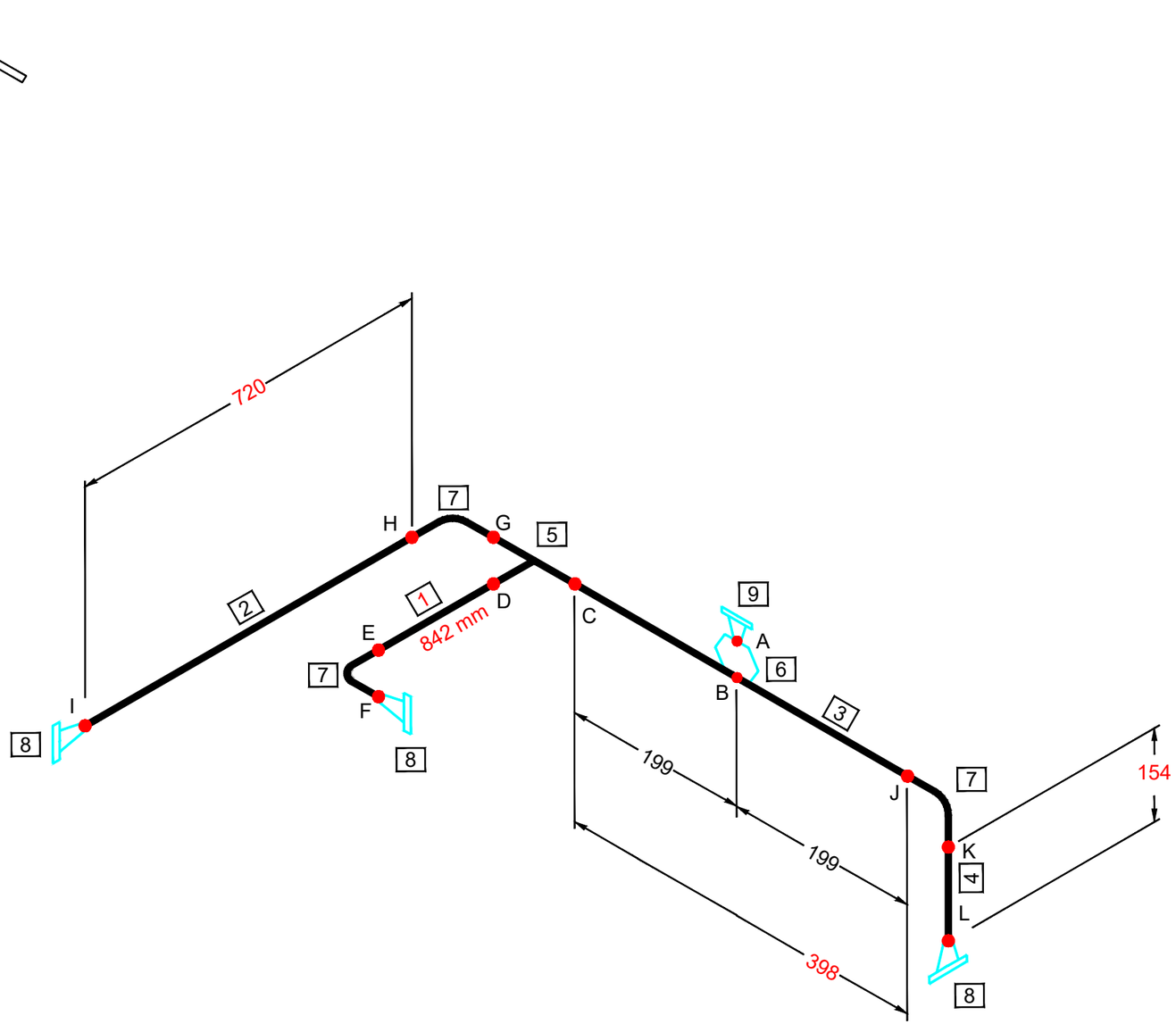
SPOOL NUMBER: 168.3-SP-CLS-725310-38H-C

REV: 0

CONTROL NUMBER: 8

ARC RESOURCES LTD.

PURNELL



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2020	1	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	842 mm
AG2020	2	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	720 mm
AG2020	3	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	398 mm
AG2020	4	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	154 mm
			FITTINGS	
PK808	5	8X8	TEE, STR., SCH. 80, BW, A420-WPL6, NACE	1
51413	6	8X2	WOL, SCH. 160, BW, A350-LF2 CL1, NACE	1
10VPY	7	8	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	3
			FLANGES	
2304	8	8	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	3
0618	9	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	1
			GASKETS	
	10	8	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	3
	11	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	12	1.1/8	(12) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (200.000 MM LG)	3
	13	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	DJM	XF-66			DM
B	2"	LET	DJM		MT-10		DM
C	8"	BW	DJM	XF-57			DM
D	8"	BW	DJM	XF-63			DM
E	8"	BW	DJM	XF-64			DM
F	8"	BW	DJM	XF-65			DM
G	8"	BW	DJM	XF-60			DM
H	8"	BW	DJM	XF-61			DM
I	8"	BW	DJM	XF-62			DM
J	8"	BW	DJM	XF-58			DM
K	8"	BW	DJM	XF-57			DM
L	8"	BW	DJM	XF-56			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

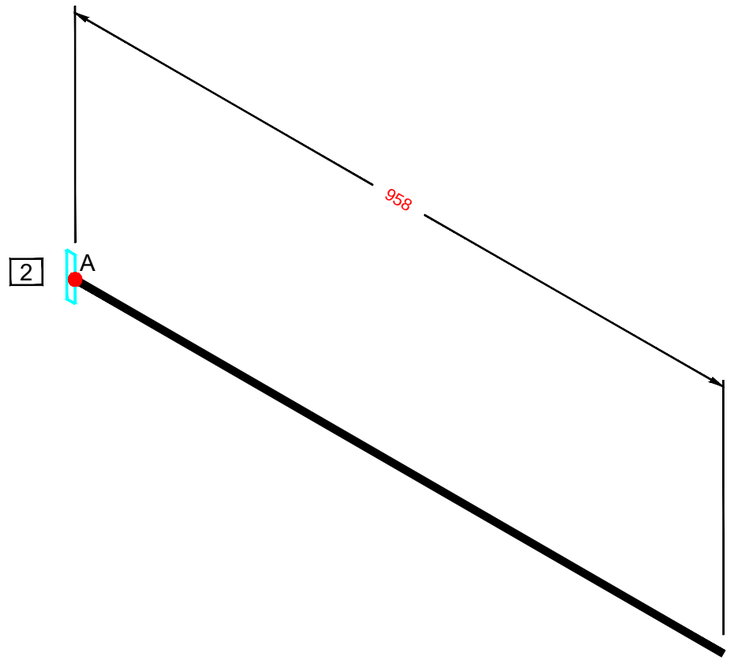
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 168.3-NG-CLS-725311-38H-A

REV: 0

CONTROL NUMBER: 9

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2020	1	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	958 mm
			VALVES / IN-LINE ITEMS	
LH8438	2	8	FLG, BLIND, RF, ASME 600, ASTM A350-LF2-CL1	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	8"	LET	DJM		MT-3		DM

DUMMY LEG

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	N/A	-45°C to 38°C	WELD DIA / INCH		X-RAY	N/A	INSULATION SPEC	N/A
MIN. TEST:	N/A	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME & PAINT
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 168.3-NG-CLS-725311-38H-B

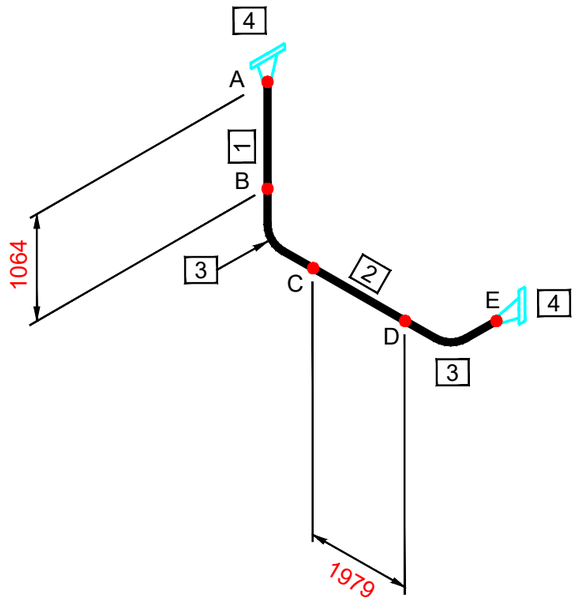
REV: 0

CONTROL NUMBER: 10

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	1064 mm
AG2073	2	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	1979 mm
			FITTINGS	
10VPY, 65C01M032-1	3	2	ELBOW, 90 LR, SCH. 160, BW, A420-WPL6, NACE	2
			FLANGES	
0618	4	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			GASKETS	
	5	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	6	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-28			DM
B	2"	BW	GC	XF-27			DM
C	2"	BW	GC	XF-26			DM
D	2"	BW	GC	XF-25			DM
E	2"	BW	GC	XF-24		XF-24	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	5	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	10.0	X-RAY	100%	INSULATION SPEC	25H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-730218-25H-A

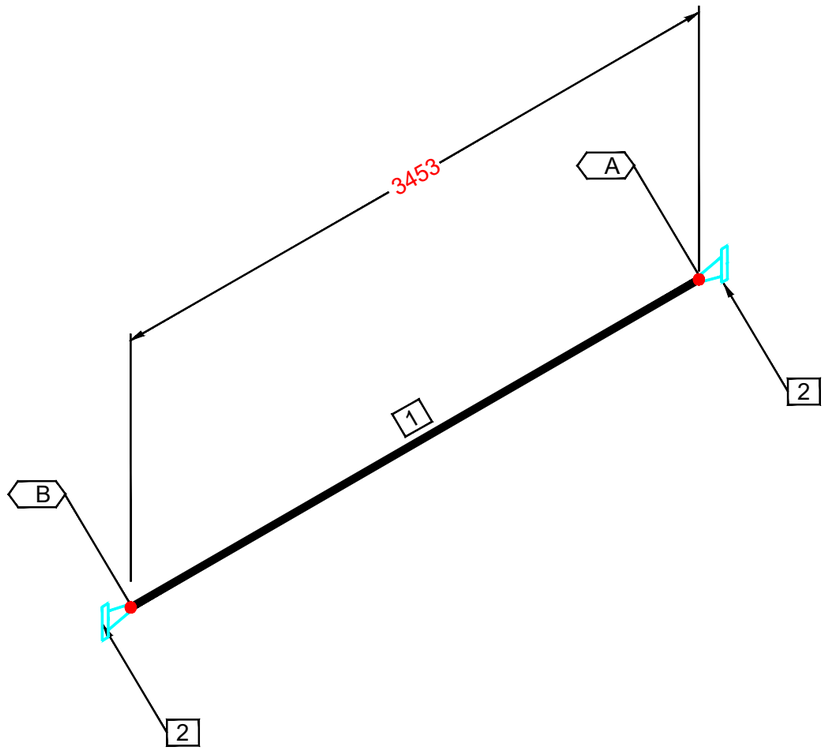
REV: 0

CONTROL NUMBER 11

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	3453 mm
			FLANGES	
0618	2	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			BOLTS	
	3	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-44			DM
B	2"	BW	GC	XF-43		XF-43	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	2	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	4.0	X-RAY	100%	INSULATION SPEC	25 H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

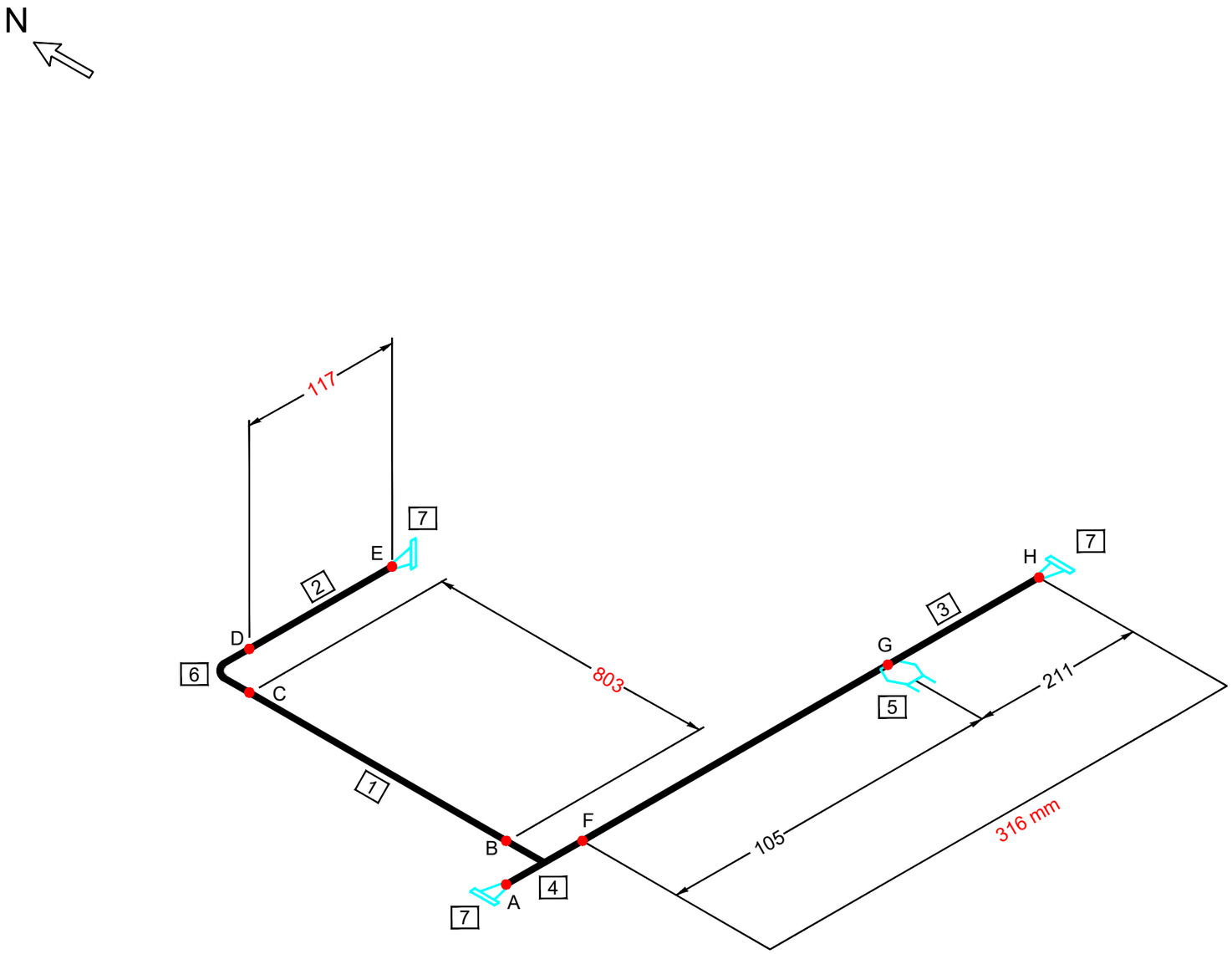
ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-730218-25H-B

REV: 0

CONTROL NUMBER: 12



BILL OF MATERIALS

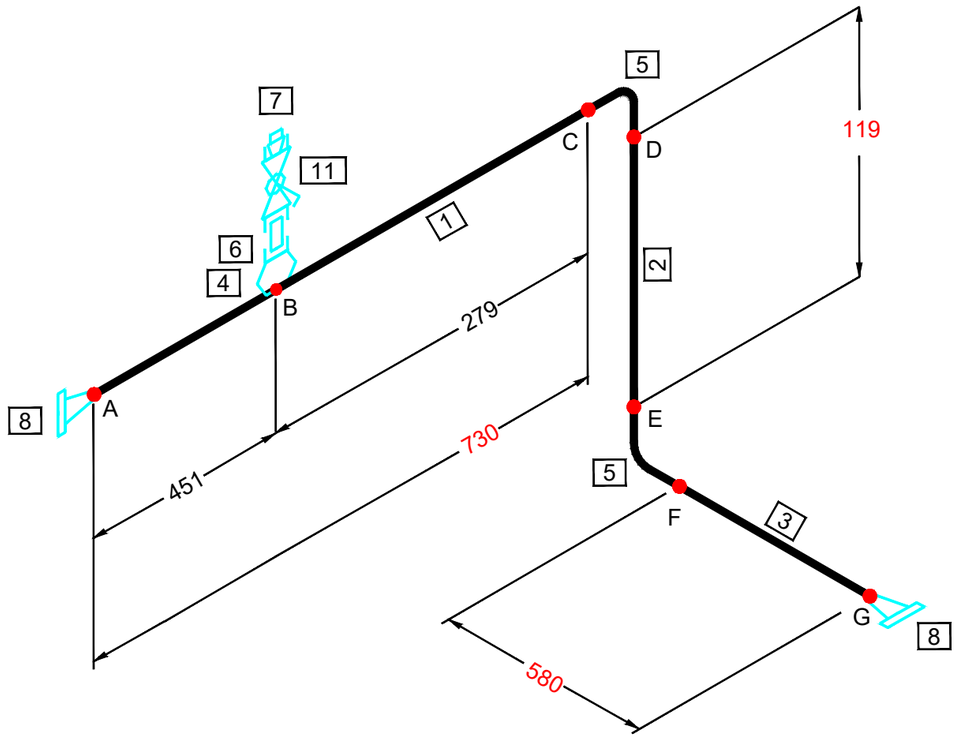
HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	803 mm
AG2073	2	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	117 mm
AG2073	3	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	316 mm
			FITTINGS	
65E11M220	4	2X2	TEE, STR., SCH. 160, BW, A420-WPL6, NACE	1
51597	5	2X1	TOL, CL6000, A350-LF2 CL1, NACE	1
07VPY	6	2	ELBOW, 90 LR, SCH. 160, BW, A420-WPL6, NACE	1
			FLANGES	
0618	7	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	3
			GASKETS	
	8	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	2
			BOLTS	
	9	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-33			DM
B	2"	BW	GC	XF-32			DM
C	2"	BW	GC	XF-31			DM
D	2"	BW	GC	XF-30		XF-30	DM
E	2"	BW	GC	XF-29			DM
F	2"	BW	GC	XF-34			DM
G	1"	LET	GC		MT-5		DM
H	2"	BW	GC	XF-35			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS		ARC RESOURCES LTD.					
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	8	M.P.I.	100%	PIPING SPEC	CLS	DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	15.0	X-RAY	100%	INSULATION SPEC	25H	WELDER:		DATE:		ZONE:	KAKWA
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME	Q.C.		DATE:		LSD:	10-16-062-04 W6M
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A	SPOOL NUMBER:	60.3-NG-CLS-730218-25H-C			REV: 0	CONTROL NUMBER 13
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3						

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2172	1	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	730 mm
AG2172	2	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	119 mm
AG2172	3	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	580 mm
			FITTINGS	
UXRN	4	12X1	TOL, CL6000, A350-LF2 CL1, NACE	1
447A	5	12	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	2
I66	6	1	NIPPLE, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	1
YB0206	7	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
			FLANGES	
LH8275	8	12	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	2
			GASKETS	
	9	12	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	2
			BOLTS	
	10	1.1/4	(20) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (225.000 MM LG)	2
			VALVES / IN-LINE ITEMS	
	11	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	12"	BW	JI	XF-83			DM
B	1"	LET	JI		MT-15		DM
C	12"	BW	JI/DF	XF-82			DM
D	12"	BW	JI	XF-81			DM
E	12"	BW	JI	XF-80			DM
F	12"	BW	JI/DF	XF-79		XF-79	DM
G	12"	BW	JI	XF-78			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	51H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 323.8-NG-CLS-730512-51H-A

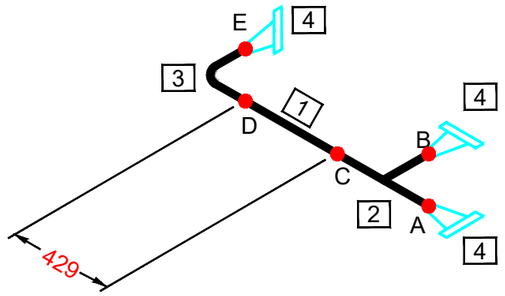
REV: 0

CONTROL NUMBER 14

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2172	1	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	429 mm
			FITTINGS	
437B	2	12X12	TEE, STR., SCH. 80, BW, A420-WPL6, NACE	1
447A	3	12	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
			FLANGES	
LH8275	4	12	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	3
			GASKETS	
	5	12	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	3
			BOLTS	
	6	1.1/4	(20) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (225.000 MM LG)	3

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	12"	BW	GC	XF-73			DM
B	12"	BW	GC	XF-74		XF-74	DM
C	12"	BW	GC	XF-75			DM
D	12"	BW	GC	XF-76			DM
E	12"	BW	GC	XF-77			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	5	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	60.0	X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

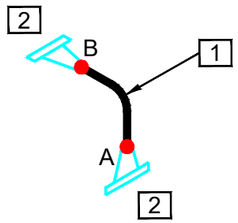
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 323.8-NG-CLS-730512-51H-B

REV: 0

CONTROL NUMBER: 15

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			FITTINGS	
B940	1	12	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
			FLANGES	
LH8275	2	12	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	2
			GASKETS	
	3	12	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	4	1.1/4	(20) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (225.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	12"	BW	GC	XF-1		XF-1	DM
B	12"	BW	GC	XF-2			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	51H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 323.8-NG-CLS-730512-51H-C

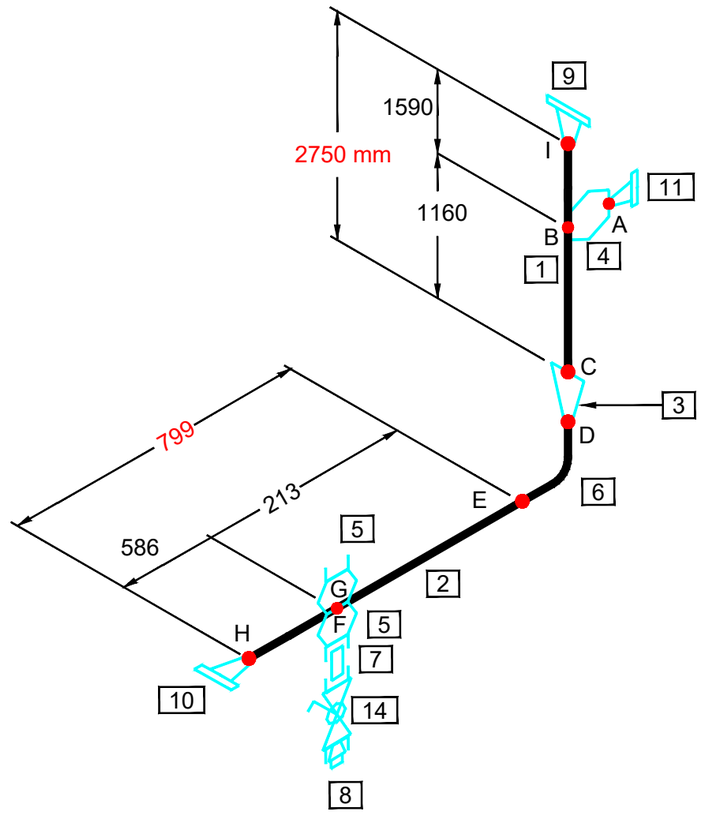
REV: 0

CONTROL NUMBER 16

ARC RESOURCES LTD.

PURNELL

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2172	1	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	2750 mm
AG2148	2	10	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	799 mm
			FITTINGS	
66A21M015	3	12X10	REDUCER, CONC., SCH. 80 x SCH. 80, BW, A420-WPL6, NACE	1
51272	4	12X2	WOL, SCH. 160, BW, A350-LF2 CL1, NACE	1
NO HT #S	5	10X1	TOL, CL6000, A350-LF2 CL1, NACE	2
533B	6	10	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
I66	7	1	NIPPLE, SCH. XXH, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	1
YB0206	8	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
			FLANGES	
LH8275	9	12	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
22/36388	10	10	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
0618	11	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	1
			GASKETS	
	12	10	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
	13	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			VALVES / IN-LINE ITEMS	
231008	14	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	DF	XF-67			DM
B	2"	LET	DF		MT-11		DM
C	12"	BW	DF	XF-71			DM
D	10"	BW	DF	XF-70			DM
E	10"	BW	DF	XF-157			DM
F	1"	LET	DF		MT-12		DM
G	1"	LET	DF		MT-13		DM
H	10"	BW	DF	XF-68		XF-68	DM
I	12"	BW	DF	XF-72			DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	51H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

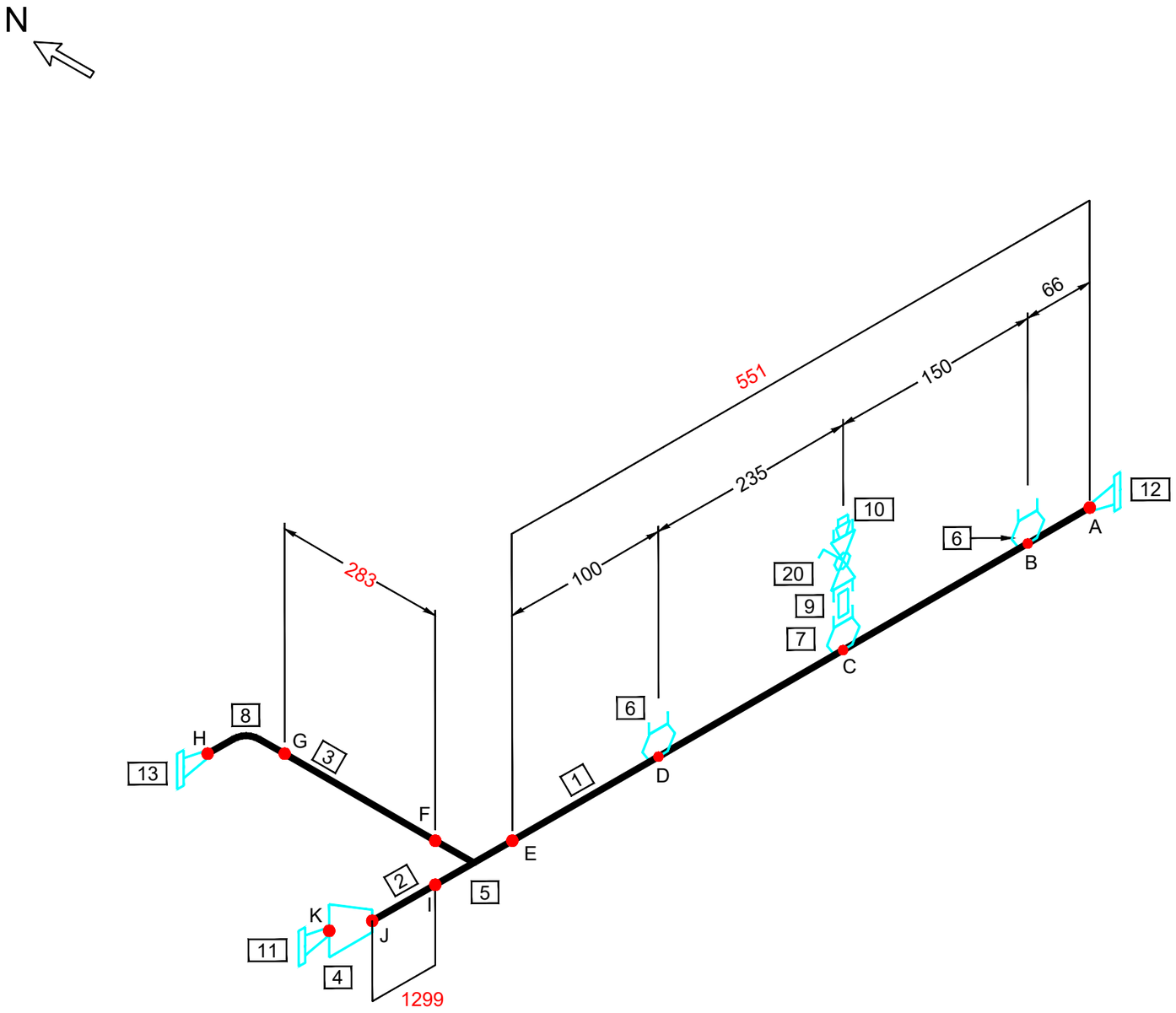
ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.05.13	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 323.8-NG-CLS-730512-51H-D

REV: 1

CONTROL NUMBER: 17



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2148	1	10	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	551 mm
AG2148	2	10	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	1299 mm
J5	3	8	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	283 mm
			FITTINGS	
EP128	4	16X10	REDUCER, ECC., SCH. 80 x SCH. 80, BW, A420-WPL6, NACE	1
TLC0035081	5	10X8	TEE, RED., SCH. 80 LE x SCH. 80 SE, A420-WPL6, NACE	1
51597	6	10X1	TOL, CL6000, A350-LF2 CL1, NACE	2
AVDY	7	10X1/2	TOL, CL6000, A350-LF2 CL1, NACE	1
C362	8	8	ELBOW, 90 LR, SCH. 80, BW, A420-WPL6, NACE	1
I64	9	1/2	NIPPLE, SCH. XXS, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	1
YB0150	10	1/2	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
			FLANGES	
10WD8	11	16	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
134005	12	10	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
D12309367	13	8	FLANGE, WN, CL600, RF, SCH. 80, A350-LF2 CL1, NACE	1
			GASKETS	
	14	16	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
	15	10	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
	16	8	GASKET, CL600, FLEXITALLIC ?FLEXICARB? CGI, 316 SS WINDINGS/IR, CS OR, OR APPROVED EQUAL	1
			VALVES / IN-LINE ITEMS	
231106	20	1/2	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	10"	BW	JI	XF-50			DM
B	1"	LET	JI		MT-6		DM
C	1/2"	LET	JI		MT-7		DM
D	1"	LET	JI		MT-8		DM
E	10"	BW	CS	XF-154			DM
F	8"	BW	CS	XF-153		XF-153	DM
G	8"	BW	CS	XF-152			DM
H	8"	BW	CS	XF-151			DM
I	10"	BW	CS	XF-155			DM
J	10"	BW	CS	XF-156			DM
K	16"	BW	JI	XF-48		XF-48	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	51H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.05.13	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

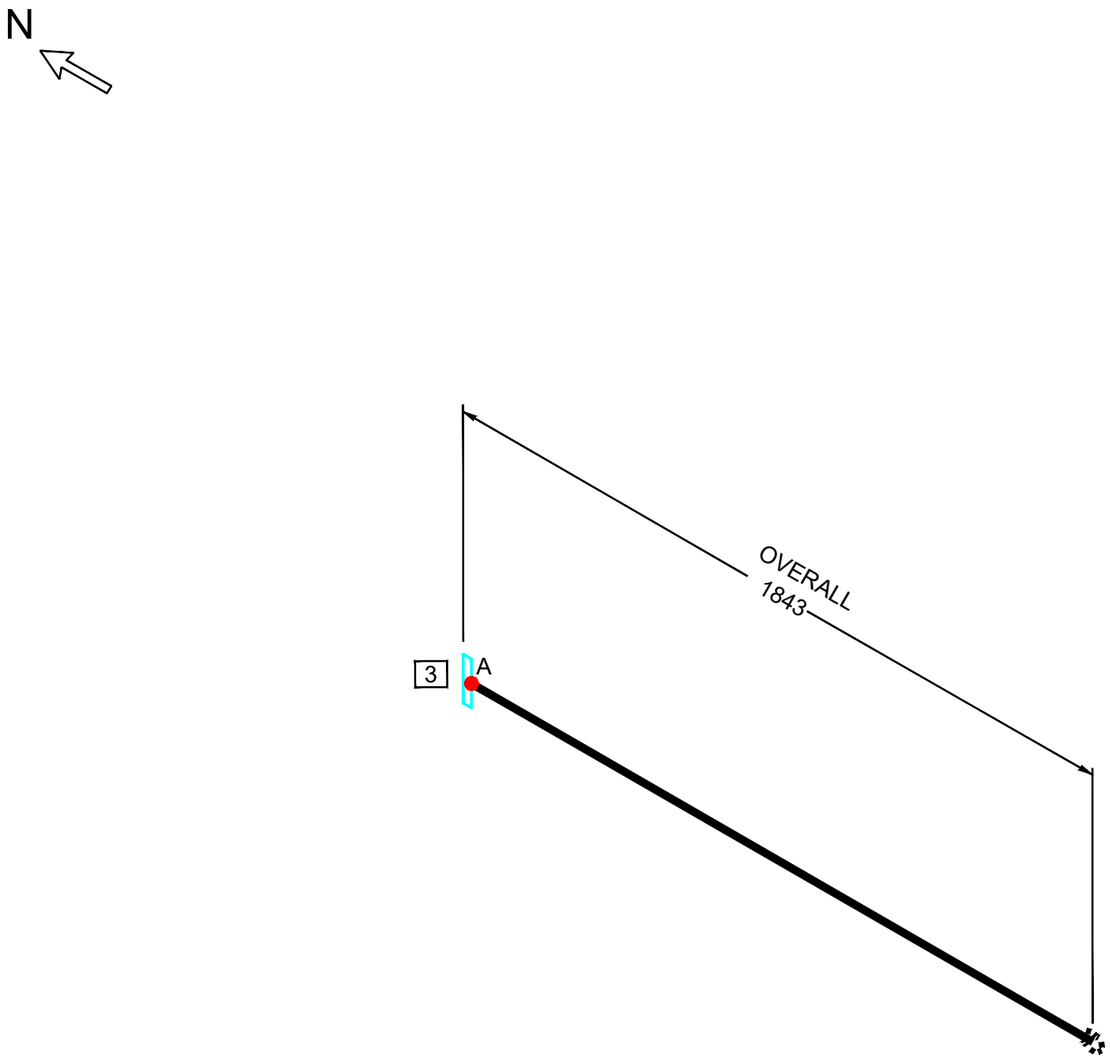
SPOOL NUMBER: 323.8-NG-CLS-730512-51H-E

REV: 2

CONTROL NUMBER: 18

ARC RESOURCES LTD.

PURNELL



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2172	1	12	PIPE, BE, SCH. 80, SMLS, A333-6, NACE	1770 mm
			GASKETS	
	2	12	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			VALVES / IN-LINE ITEMS	
LH8420	3	12	FLANGE, BLIND, CL600, RF, A350-LF2 CL1, NACE	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	12"	LET	DJM		MT-14		DM

DUMMY LEG

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	N/A	-45°C to 38°C	WELD DIA / INCH		X-RAY	N/A	INSULATION SPEC	N/A
MIN. TEST:	N/A	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME & PAINT
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

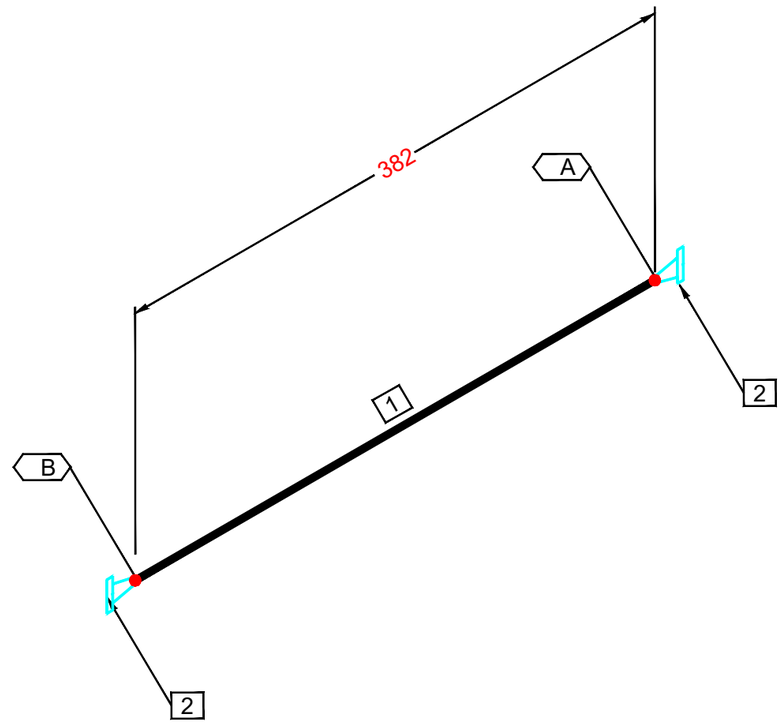
DRAFTER:	SEC	DATE:	2024.04.15	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 323.8-NG-CLS-730512-51H-F

REV: 0

CONTROL NUMBER: 19

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
AG2073	1	2	PIPE, BE, SCH. 160, SMLS, A333-6, NACE	382 mm
			FLANGES	
0618	2	2	FLANGE, WN, CL600, RF, SCH. 160, A350-LF2 CL1, NACE	2
			GASKETS	
	3	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	4	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
A	2"	BW	GC	XF-38			DM
B	2"	BW	GC	XF-37		XF-37	DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS	2	M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH	4.0	X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.05.13	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 60.3-NG-CLS-730513-25H

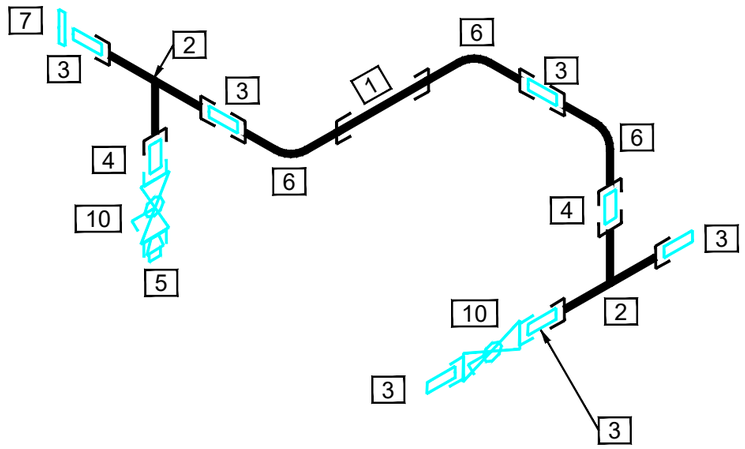
ARC RESOURCES LTD.

PURNELL

REV: 1

CONTROL NUMBER 20

N



BILL OF MATERIALS

HEAT #	ITM #	SIZE	DESCRIPTION	QTY
			PIPE	
T781986	1	1	PIPE, TBE, SCH. XXH, SMLS, A333-6, NACE (205.176 MM LG)	205 mm
			FITTINGS	
YB0124	2	1X1	TEE, STR., CL6000, THRD, A350-LF2 CL1, NACE	2
I66	3	1	NIPPLE, SCH. XXH, SMLS, A333-6, NACE, 76mm LG, TBE (76.000 MM LG)	6
I66	4	1	NIPPLE, SCH. XXH, SMLS, A333-6, NACE, 100mm LG, TBE (100.000 MM LG)	2
YB0206	5	1	PLUG, HEX HEAD, THRD, A350-LF2 CL1, NACE	1
BESN	6	1	ELBOW, 90 LR, CL6000, THRD, A350-LF2 CL 1, FS, NACE	3
			FLANGES	
BT0170G	7	2X1	FLANGE, RED, HUB, THRD, CL600, RF, A350-LF2 CL1, NACE	1
			GASKETS	
	8	2	GASKET, ASME 600 RF, 3.2mm, FLEXITALLIC "FLEXICARB" CGI, 304L OR 316L SS WINDINGS AND INNER RING, FLEXIBLE GRAPHITE FILLER, ASME B16.20	1
			BOLTS	
	9	5/8	(8) STUD BOLTS, ASTM A193-B7M c/w (2 PER STUD) ASTM A194-2HM HEAVY SEMI FINISHED HEX NUTS (115.000 MM LG)	1
			VALVES / IN-LINE ITEMS	
231008	10	1	VALVE, BALL, 3000 PSI, FNTP, RP, A350-LF2-CL1, 316 SS TRIM, RTFE SEATS, 3 PIECE 4 BOLT DESIGN, FLOATING, LEAVER OP., NACE, 30BVTT4H	2

WELD INFORMATION

#	SIZE	TYPE	ID	X-RAY	MPI	HT	VT
							DM

TESTING			SPOOL DATA		NDT TESTING		SPECIFICATIONS	
CONDITION:	PRESSURE	TEMP.	TOTAL NO. WELDS		M.P.I.	100%	PIPING SPEC	CLS
DESIGN:	10,205 kPa	-45°C to 38°C	WELD DIA / INCH		X-RAY	100%	INSULATION SPEC	38H
MIN. TEST:	15,307 kPa	AMBIENT	SURFACE AREA		PWHT	N/A	PAINTING SPEC	PRIME
TEST MEDIUM:	WATER		WPS PES-1N (REV.2)		GAP SHOT	10%	TRACING SPEC	N/A
SERVICE:	NFS				HARDNESS	YES	CODE:	ASME B31.3

ARC RESOURCES LTD.

DRAFTER:	SEC	DATE:	2024.04.18	PROJ #:	2024-062A
WELDER:		DATE:		ZONE:	KAKWA
Q.C.		DATE:		LSD:	10-16-062-04 W6M

SPOOL NUMBER: 33.4-NG-CLS-730219-25H

REV: 0

CONTROL NUMBER: 23

ARC RESOURCES LTD.

PURNELL


KEEP TIGHT
IN EXPLOSIVE ATMOSPHERE
DO NOT CIRCUIT ALIVE
PSI


Falcon
Electronic DeadWeight
Menu
+
Save
DDW-007


C9808
5 TON


5 TON
C9610-A

